# LAP Webapp Feature Review

Date: 2026-02-27
Source: Repository inspection of ‘lap-website‘ (routes, app navigation, workflow definitions, API handlers, and E2E coverage).

Companion export: ‘docs/reports/webapp-feature-review.pdf‘

## Executive Summary

The webapp is organized into five product surfaces:
1. **Marketing and acquisition site** (‘/(marketing)‘ routes).
2. **Authentication and account onboarding** (‘/(auth)‘ routes).
3. **Free-report funnel** (‘/(free-report)‘ routes).
4. **Core telemetry application** (‘/app/*‘ routes).
5. **Backend-facing web APIs** (‘/api/*‘ routes) for leads, conversion tracking, billing, and public status.

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## 1) Marketing + Conversion Features

### Public marketing pages
- Landing and product messaging (‘/‘, ‘/product‘, ‘/pricing‘, ‘/demo‘, ‘/book‘).
- Persona pages (‘/drivers‘, ‘/circuits‘, ‘/teams‘, ‘/endurance‘).
- Trust and policy pages (‘/security‘, ‘/status‘, ‘/privacy‘, ‘/terms‘, ‘/dpa‘).
- Proof/SEO pages (‘/customers‘, ‘/insights‘, ‘/insights/[slug]‘).
- Lead-capture support pages (‘/contact‘, ‘/waitlist‘, ‘/thank-you/[form]‘, ‘/billing‘).

### Navigation model
- **Top navigation**: Product, Pricing, Drivers, Circuits, Teams, Endurance, Customers, Security, Insights.
- **Footer navigation**: Contact, Book, Waitlist, Billing, Status, Privacy, Terms, DPA.

### Conversion mechanics
- Lead forms are validated and forwarded to CRM/email webhook targets.
- Attribution + session IDs can be carried from cookies into lead payloads.
- Server-side conversion events are emitted for analytics pipelines.

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## 2) Auth + User Access Features

- ‘/login‘ for sign-in.
- ‘/register‘ for account creation.
- ‘/verify-email‘ flow for email verification.
- Billing API paths enforce auth for sensitive actions (portal and plan switch), with optional guest mode based on billing auth policy.

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## 3) Free Report Funnel Features

- ‘/free-report/analyze‘: upload and trigger a free telemetry analysis flow.
- ‘/free-report/report/[id]‘: view generated free report result page by report/session id.

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## 4) Core App Feature Review (‘/app‘)

The in-product sidebar exposes these primary modules:
- Dashboard
- Sessions
- Tracks
- Drivers
- Analyze
- Compare
- Strategy
- Leaderboard
- Analytics
- Karts
- Weekends
- Annotations
- Dev Plans
- Exports
- Data Health

### 4.1 Dashboard
- Portfolio-level telemetry overview with key stats:
  - total sessions
  - personal best time
  - average score
  - total drivers
- Sparkline trend cards (recent score trend and lap-time trend).
- Recent sessions data table with click-through into session detail.

### 4.2 Session Intelligence
- Session list (‘/app/sessions‘) and session detail (‘/app/sessions/[id]‘).
- Corner-level drilldown (‘/app/sessions/[id]/corners/[cornerId]‘).
- Workflow module definitions reference dedicated views across overview/corners/timeline/analytics/lap-data style analysis surfaces.

### 4.3 Track Intelligence
- Track directory (‘/app/tracks‘).
- Track detail pages (‘/app/tracks/[id]‘).
- Track evolution history view (‘/app/tracks/[id]/evolution‘).

### 4.4 Driver Management + Progression

- Driver roster (‘/app/drivers‘).
- Driver profile (‘/app/drivers/[id]‘).
- Driver progression timeline (‘/app/drivers/[id]/progression‘).

### 4.5 Data Capture and Analysis
- New analysis workflow (‘/app/analyze‘) for ingest/processing.
- Setup tracker (‘/app/setup-tracker‘) for setup configuration records.
- Annotation workspace (‘/app/annotations‘) for contextual notes.

### 4.6 Comparison Suite
- Comparison home (‘/app/compare‘).
- Session-vs-session comparison (‘/app/compare/sessions‘).
- Driver-vs-driver comparison (‘/app/compare/drivers‘).

### 4.7 Strategy Control Room
- Strategy index (‘/app/race-strategy‘).
- Create strategy plan (‘/app/race-strategy/new‘).
- Strategy detail (‘/app/race-strategy/[id]‘).
- Live strategy mode (‘/app/race-strategy/[id]/live‘).
- Strategy summary/report (‘/app/race-strategy/[id]/summary‘).

### 4.8 Performance Ranking + Analytics
- Leaderboard (‘/app/leaderboard‘).
- Analytics hub (‘/app/analytics‘) and focused tools:
  - Performance trends (‘/app/analytics/trends‘)
  - Weather analysis (‘/app/analytics/weather‘)
  - Meteo correlation (‘/app/analytics/meteo‘)
  - Pace predictor (‘/app/analytics/pace-predictor‘)
  - Weight simulator (‘/app/analytics/weight-simulator‘)

### 4.9 Entity Management and Operations
- Kart list/create/detail (‘/app/karts‘, ‘/app/karts/new‘, ‘/app/karts/[id]‘).
- Weekend list/create/detail (‘/app/weekends‘, ‘/app/weekends/new‘, ‘/app/weekends/[id]‘).
- Export center (‘/app/exports‘).
- Data health queue/list (‘/app/data-health‘) + job detail (‘/app/data-health/[id]‘).
- Dev plans list/detail (‘/app/dev-plans‘, ‘/app/dev-plans/[id]‘).
- Visualization workspace (‘/app/visualize/[id]‘).

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## 5) API Feature Review

### Lead + conversion APIs
- ‘POST /api/leads‘
  - validates payload fields
  - enriches UTM/session attribution from cookies
  - forwards to CRM + email delivery targets
  - emits ‘lead_submitted‘ conversion event
  - returns fallback success messaging if integrations are not configured

- ‘POST /api/conversions‘
  - records server-side conversion events for specified event names

### Billing APIs
- ‘POST /api/billing/checkout‘
  - supports starter redirect, B2B contact-only mode, Stripe checkout, provider webhook fallback, and local fallback URLs
- ‘POST /api/billing/portal‘
  - authenticated billing portal access via Stripe when available, with provider/fallback options
- ‘POST /api/billing/switch‘
  - authenticated subscription plan change path via Stripe, provider webhook, or fallback record mode

### Status API
- ‘GET /api/status‘
  - pulls public operational status from status endpoints
  - gracefully returns fallback "Operational" payload when upstream endpoints are unavailable

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## 6) Workflow Capability Matrix (as declared in data model)

Declared product workflow modules include:
- Upload and Analyze
- Session Intelligence
- Comparison and Progress
- Data Management
- Visualization Workspace
- Race Strategy Control Room
- Coaching Plans
- Account and Team
- Data Rights

Most are marked **available_now**, while strategy control room and account/team are marked **enterprise_on_request**.

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## 7) QA and Test Surface (Webapp)

The webapp includes E2E specs covering key journeys:
- auth
- marketing
- upload
- sessions
- tracks
- compare
- strategy

- exports

This indicates automated coverage around both acquisition pages and core telemetry workflows.

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## 8) Notes / Limitations of This Review

- This review is code- and route-driven from the repository rather than runtime UX validation.
- Some pages are scaffolded with placeholder/demo data patterns; exact production behavior depends on backend availability and environment configuration.
- Capability tags in workflow data represent declared product intent and may differ by deployment tier.